SHETH L.U.J AND SIR M.V. COLLEGE

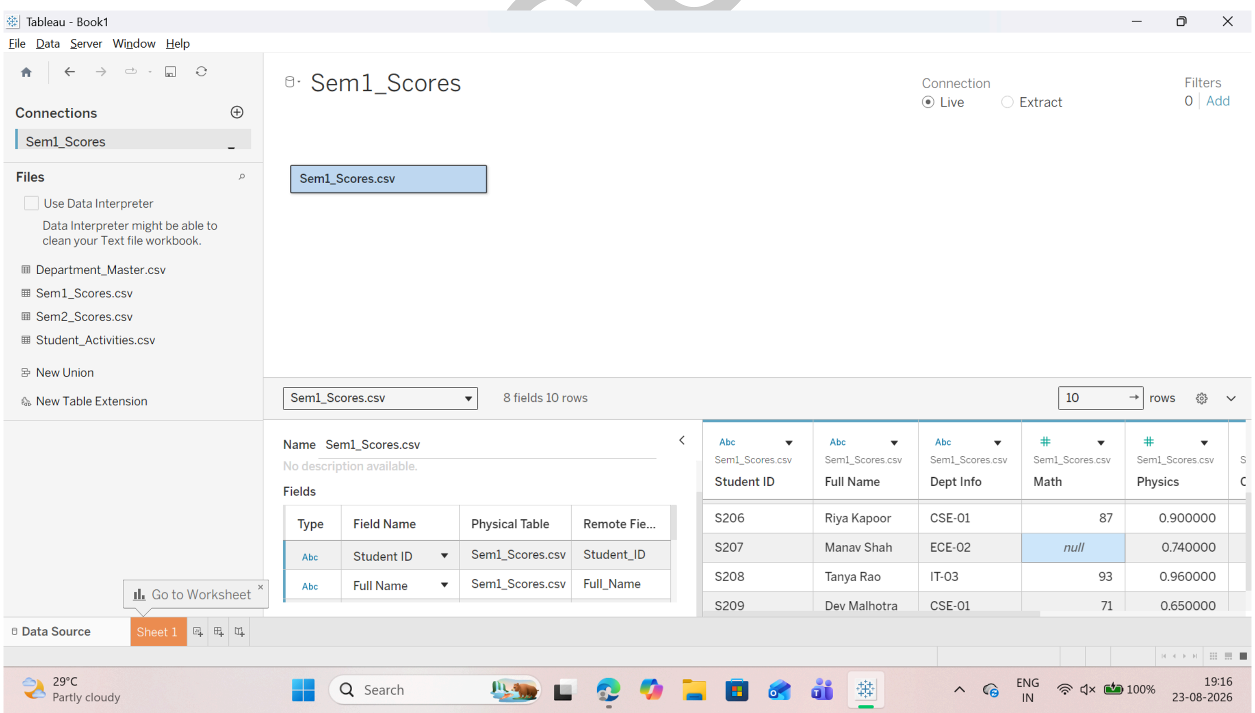
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

PRACTICAL - 9

Aim: Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data
b) Convert wide data into tall format
c) Perform union of multiple files
d) Execute cross-database joins
e) Restructure data for better visualization
f) Handle different levels of detail in datasets
g) Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data
Step 1: Load the Dataset
1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table:
o Observe missing entries (Null) in Math, Physics, and Chemistry.



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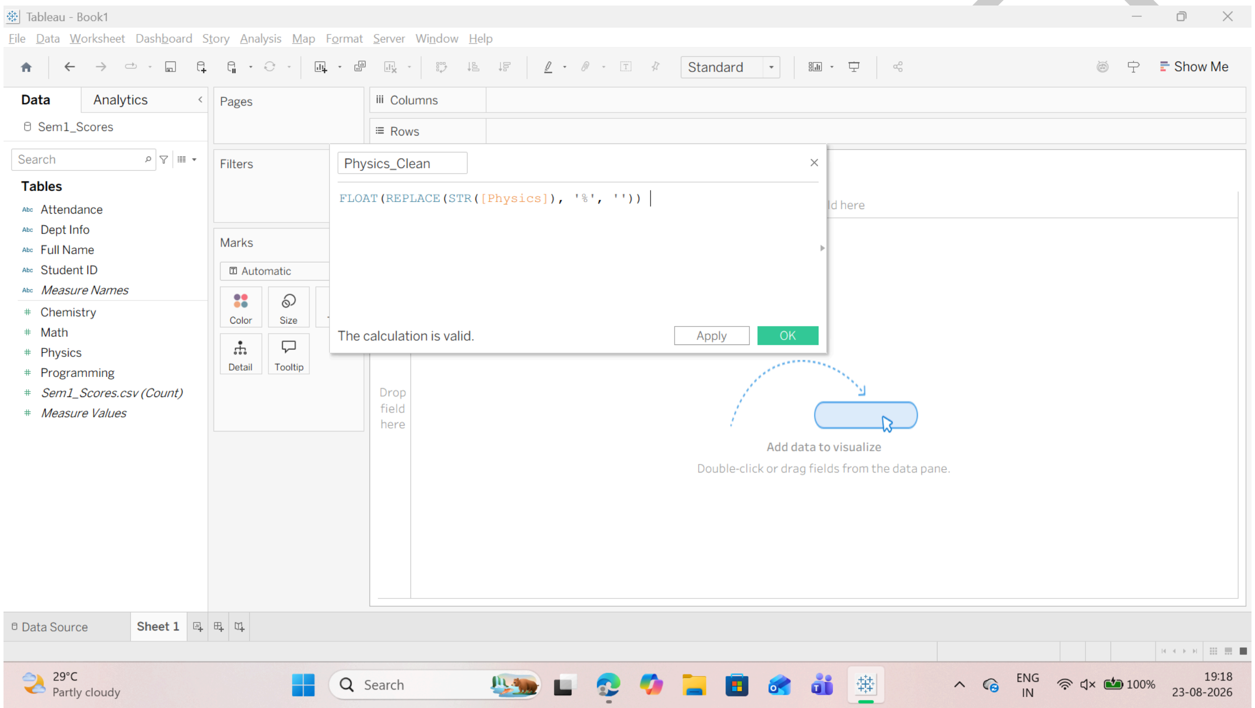
Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....
3. Create the cleaned measure for Physics:

o Field Name: Physics_Clean

o Tableau Formula:

FLOAT(REPLACE(STR([Physics]), '%', ''))



4. Create the cleaned measure for Attendance: o Right-click the Data Pane --> Create Calculated Field....

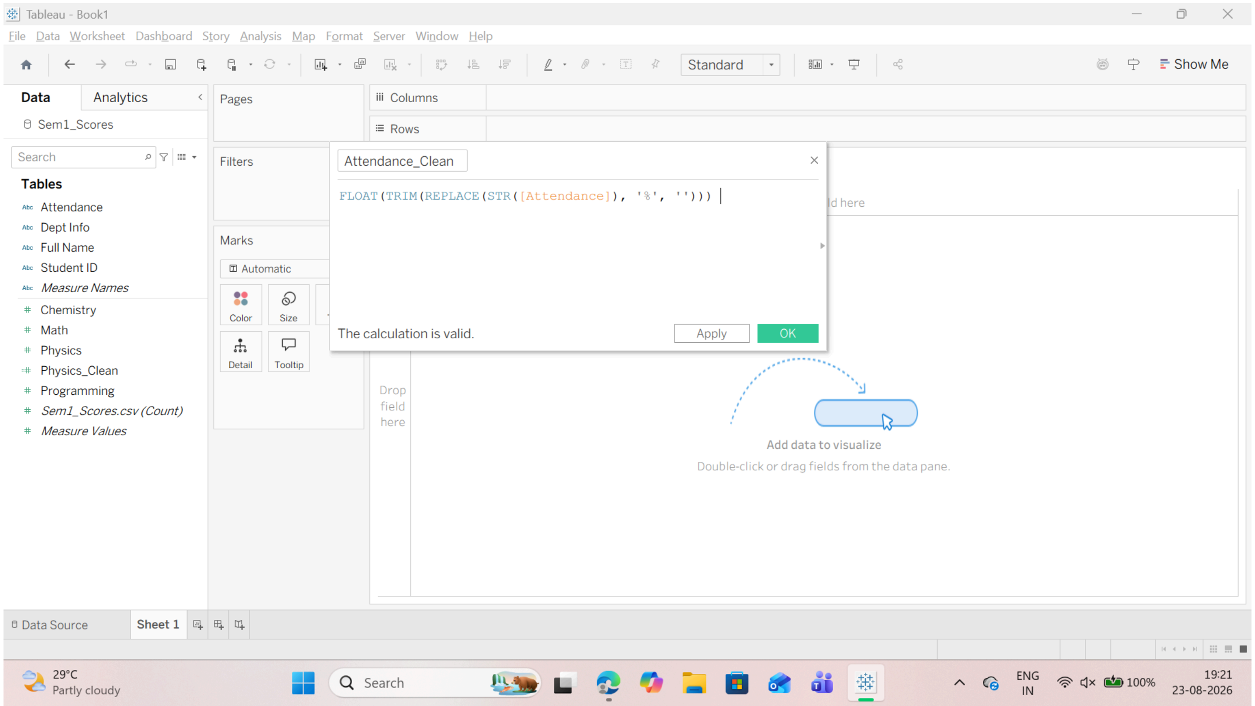
o Field Name: Attendance_Clean

o Tableau Formula:

Code snippet

FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))

o Click OK.



Step 3: Handle Missing (Null) Numeric Values via Imputation

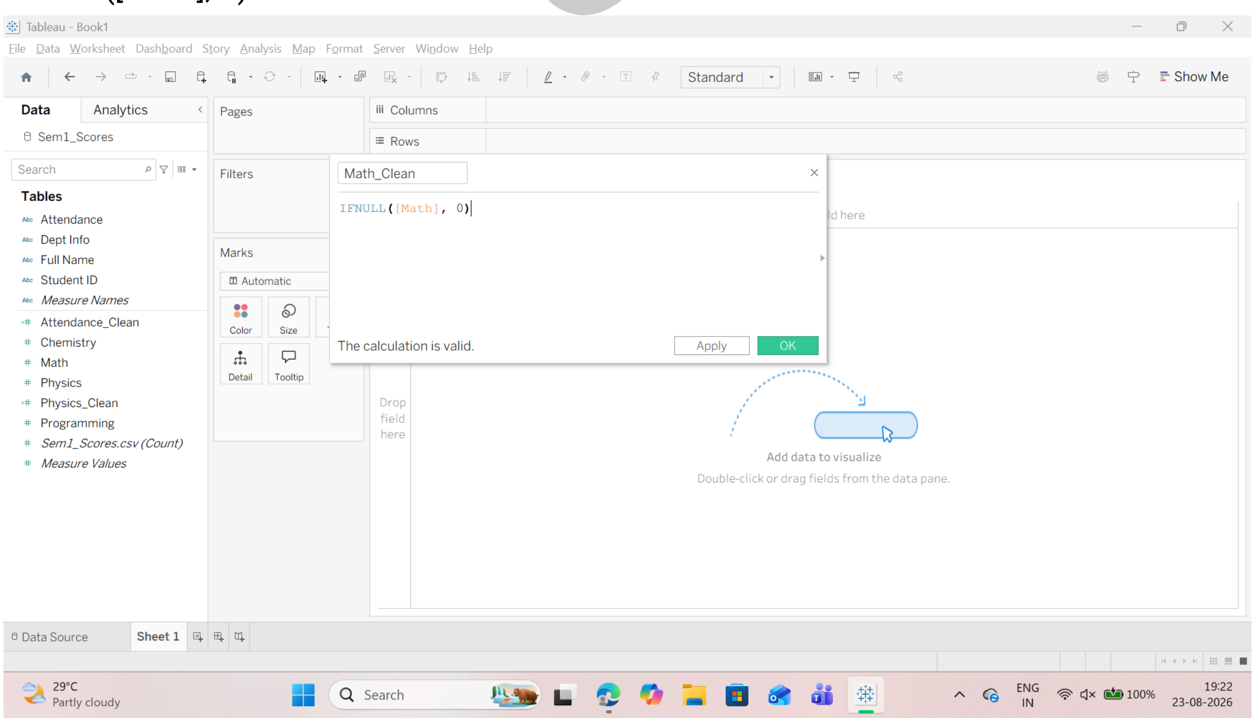
1. Handle missing entries in Math:

o Right-click the Data Pane --> Create Calculated Field....

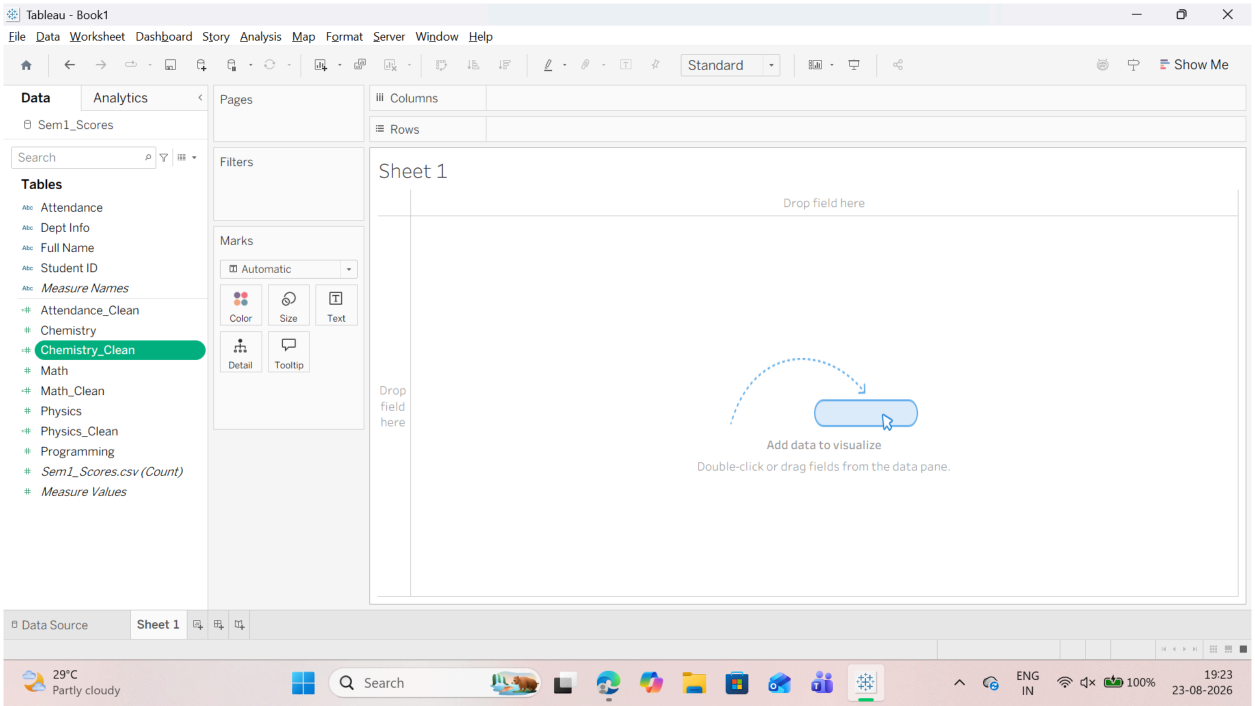
o Field Name: Math_Clean

o Tableau Formula:

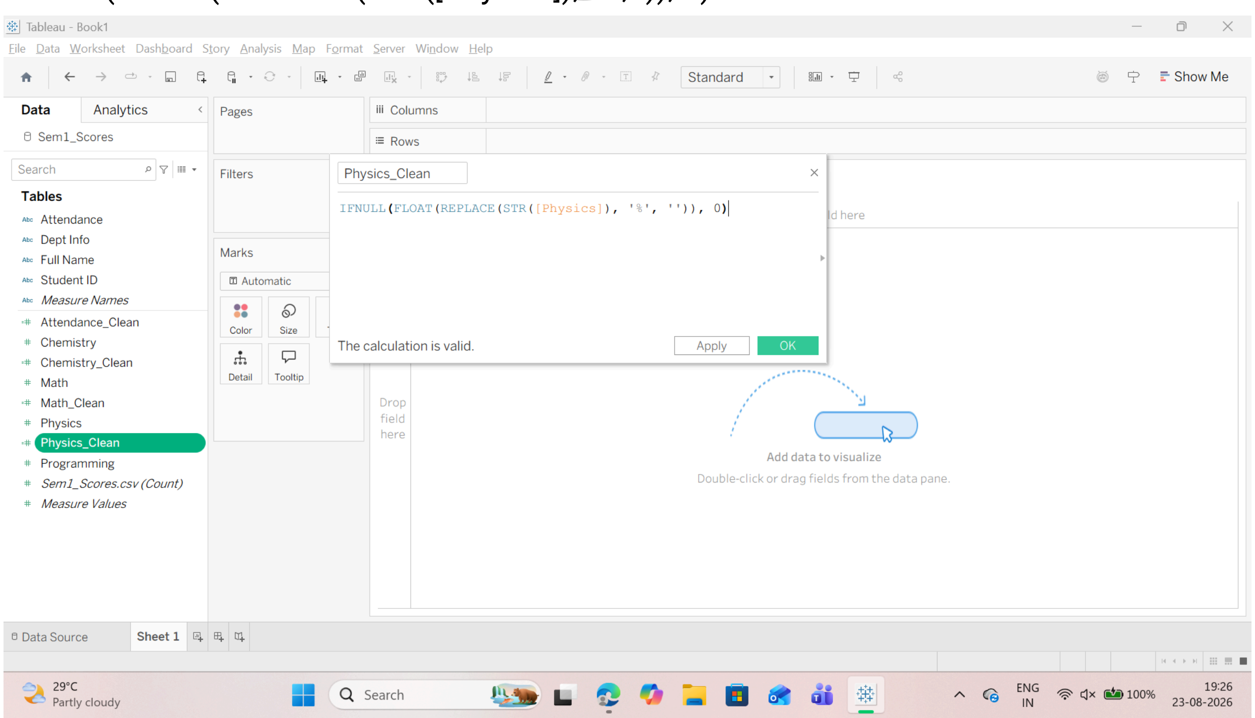
IFNULL([Math], 0)



2.Right-click Data Pane → Create Calculated Field → Name: Chemistry_Clean → Formula: IFNULL([Chemistry], 0) → Click OK.



Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)
1.S105 contains a Null value in Physics. Combine string stripping with null handling:
o Edit Physics_Clean (right-click Physics_Clean --> Edit...):
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)



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B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

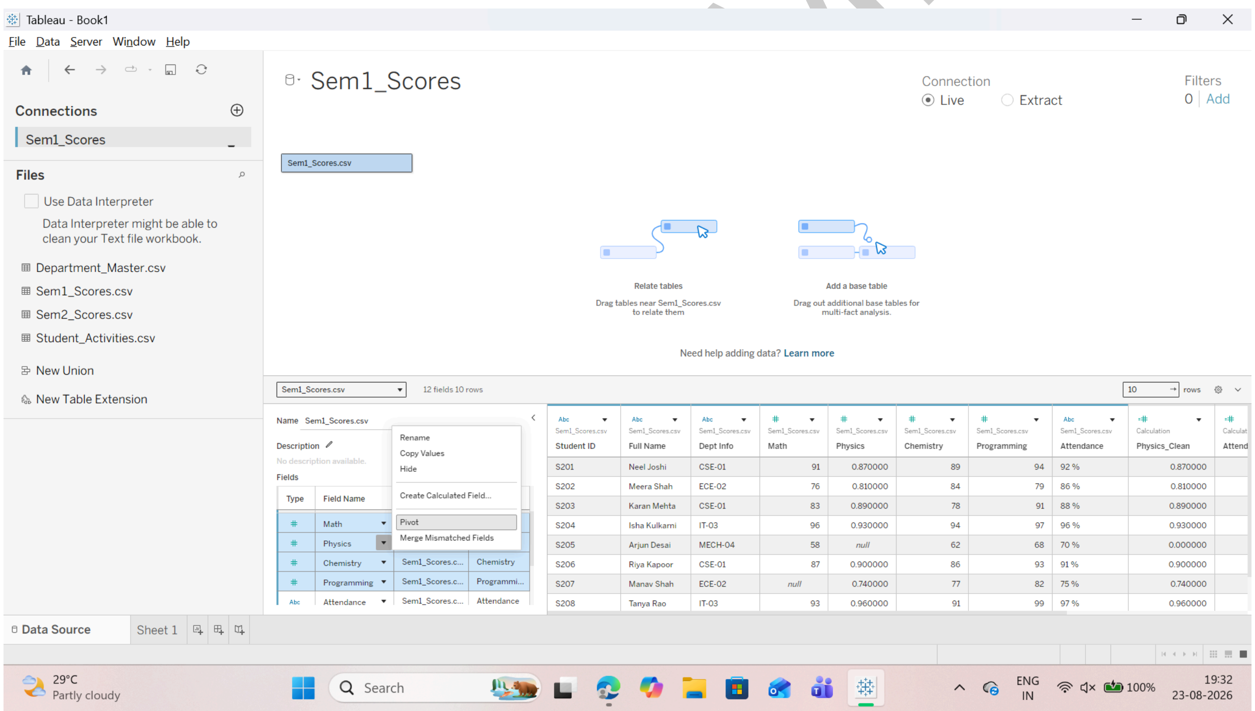
1. In Tableau, navigate to the Data Source tab at the bottom-left corner.
2. Locate the four horizontal subject score columns:

o Math o Chemistry

o Programming o Physics

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.
3. Right-click on any of the highlighted column headers.
4. Select Pivot.



Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

1. Double-click the header named Pivot Field Names and type: Subject
2. Double-click the header named Pivot Field Values and type: Score

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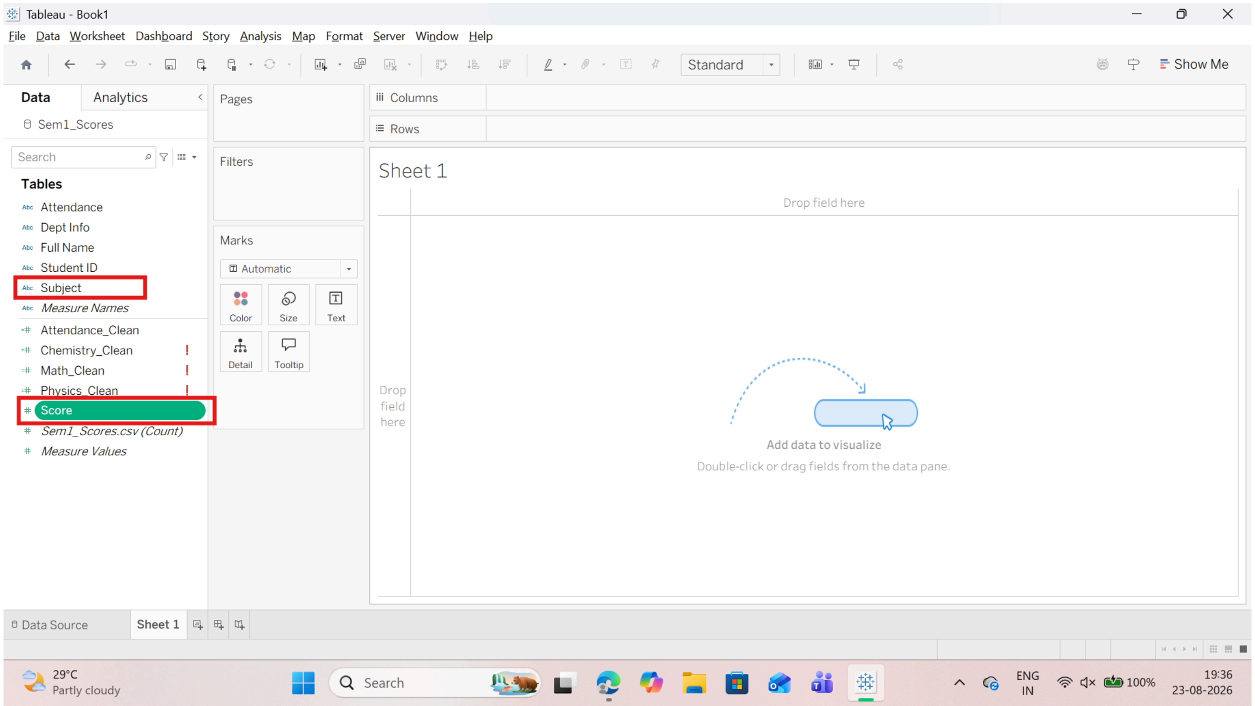
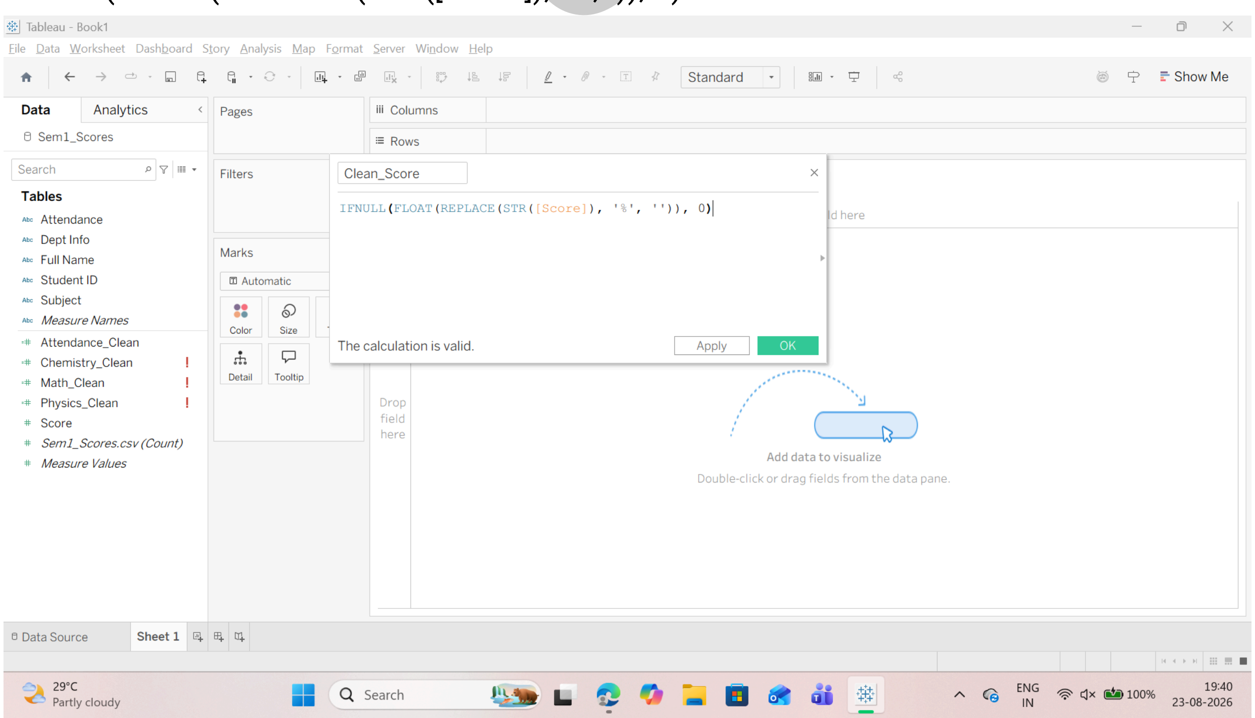


Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbol, the new combined Score column defaults to text (Abc). Convert it to numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list.
3. Alternatively, create a calculated field inside Sheet 1 named Clean_Score: IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)



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C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

Step 2: Create a Multi-File Union

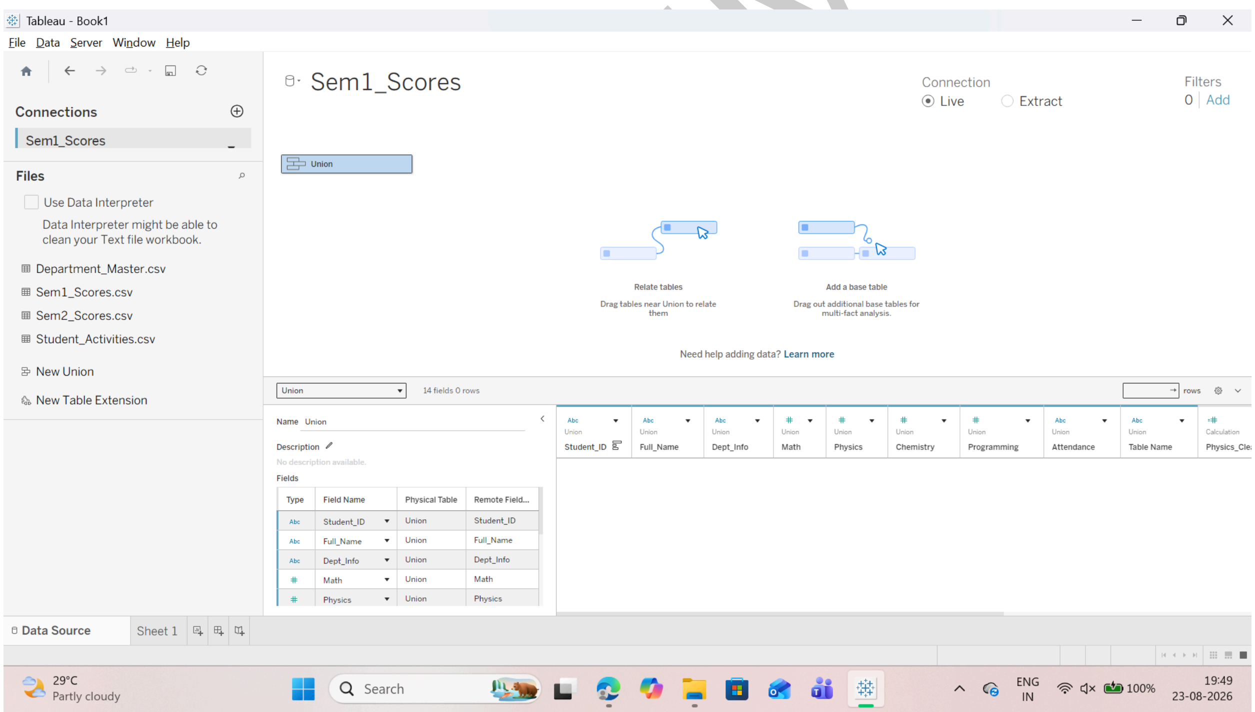
1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.
3. In the pop-up Union dialog box:

o Drag Sem1_Scores.csv from the left panel into the box.

o Drag Sem2_Scores.csv from the left panel into the same box.

o Verify that both filenames are listed in the Union box.

4. Click Apply, then click OK.
5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.



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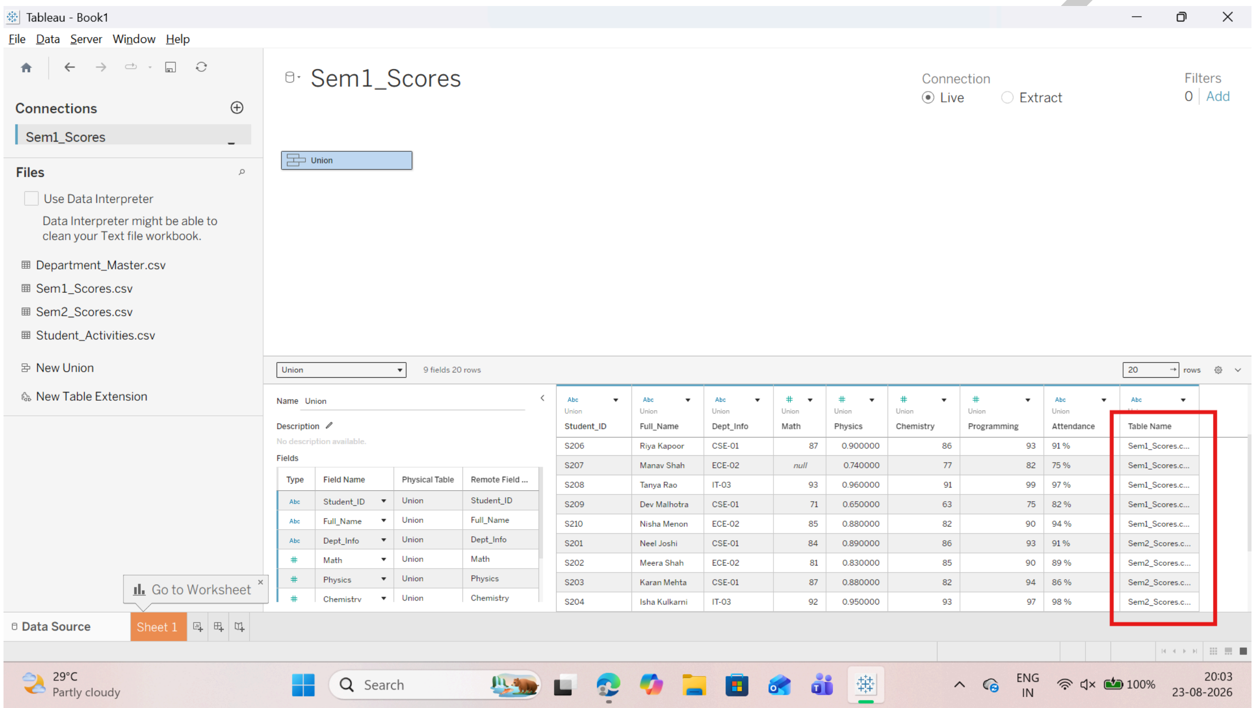
Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).
3. Scroll to the far right of the grid to verify the generated system column:

o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

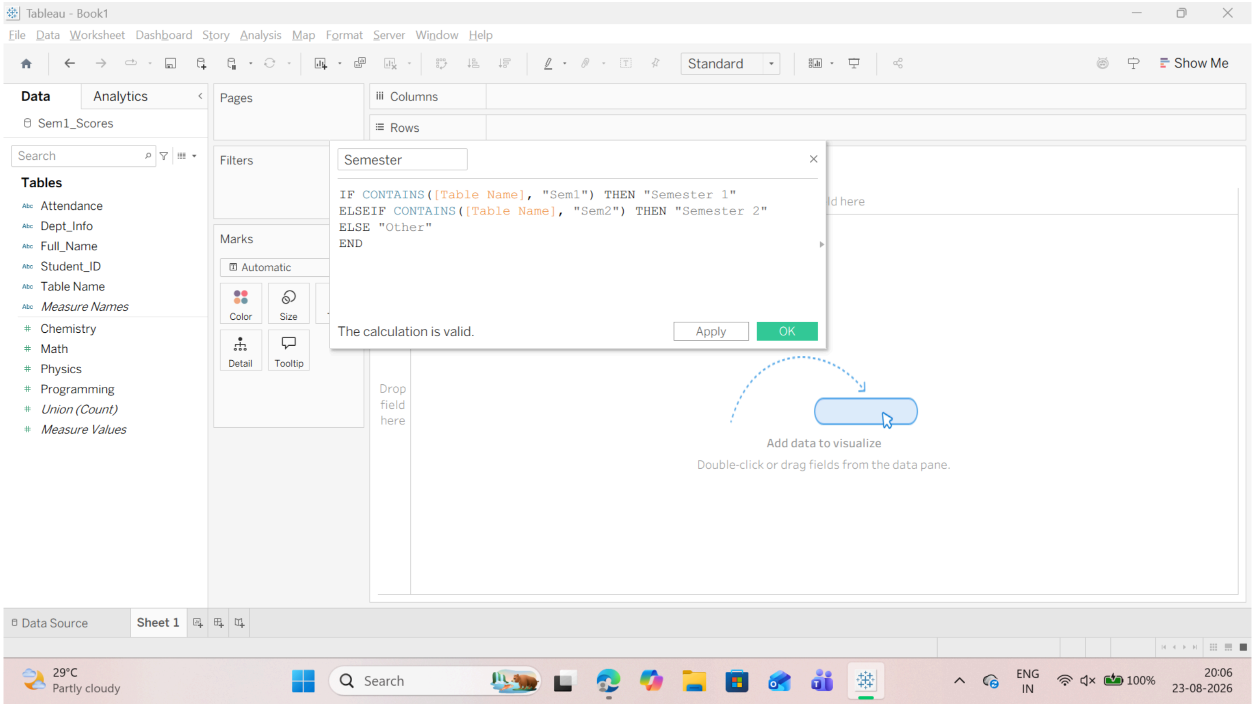


Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:

Code snippet

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```



D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas

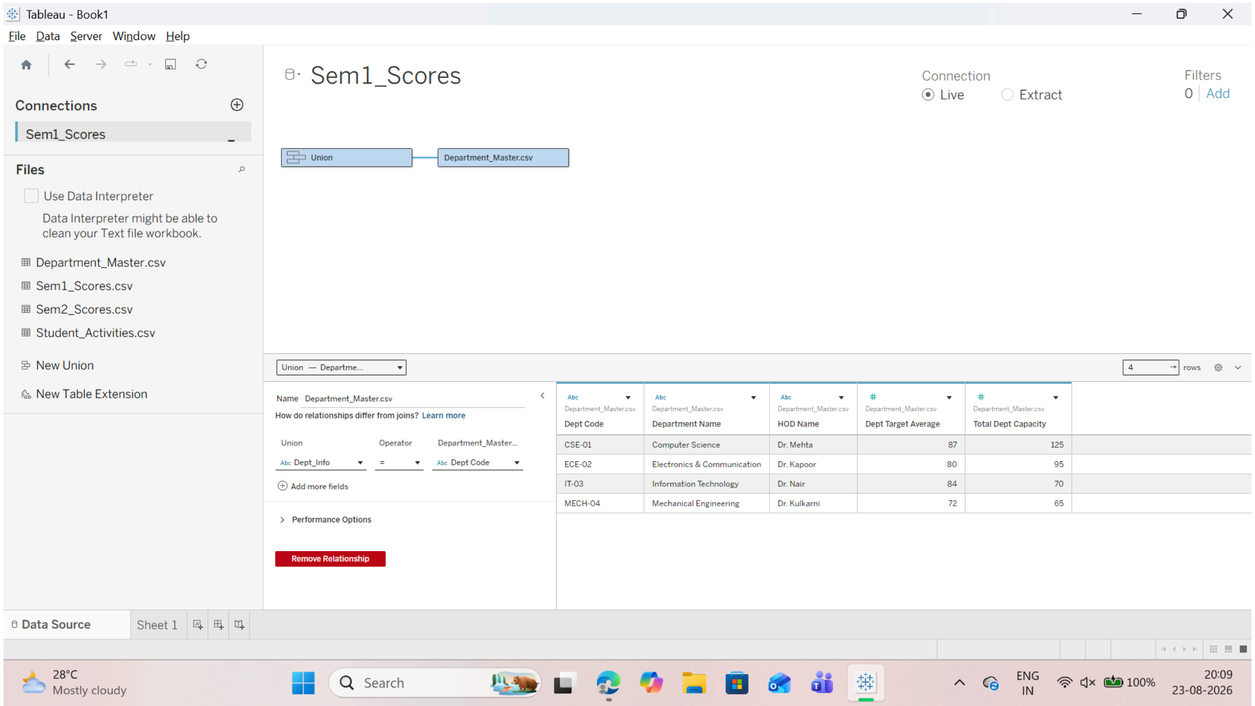
Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.
4. If the relationship dialog opens (or click the relationship line):

o Left Table Field (Union): Select Dept_Info

o Right Table Field (Department_Master): Select Dept_Code

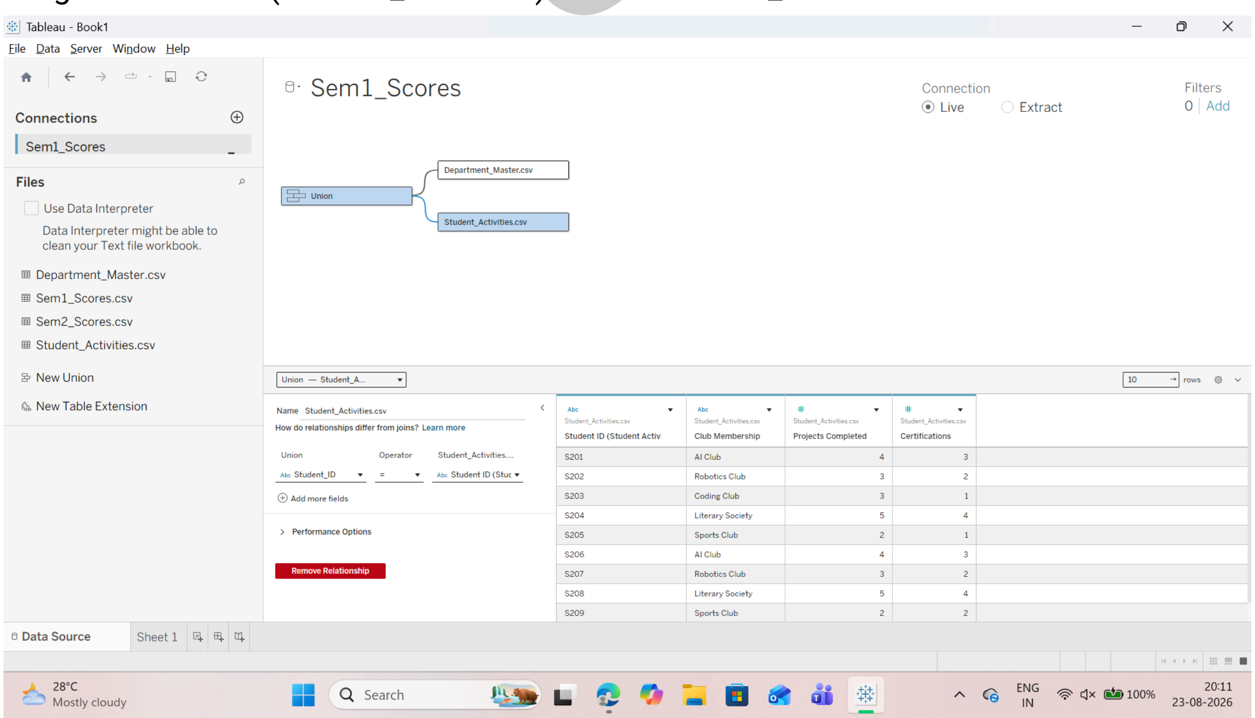
o Operator: =



Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.
3. In the relationship dialog:

o Left Table Field (Union): Select Student_ID
o Right Table Field (Student_Activities): Select Student_ID



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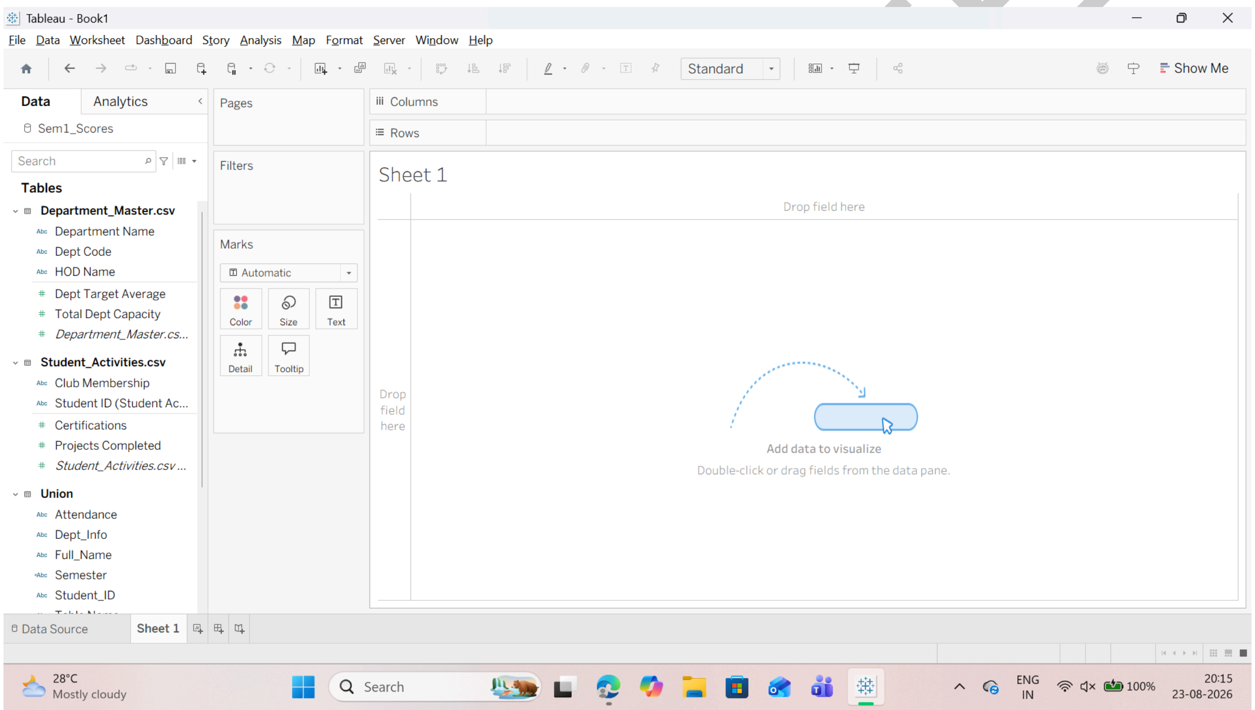
Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:

o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name

o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity

o Student_Activities: Club_Membership, Projects_Completed, Certifications



E: Restructure Data for Better Visualization

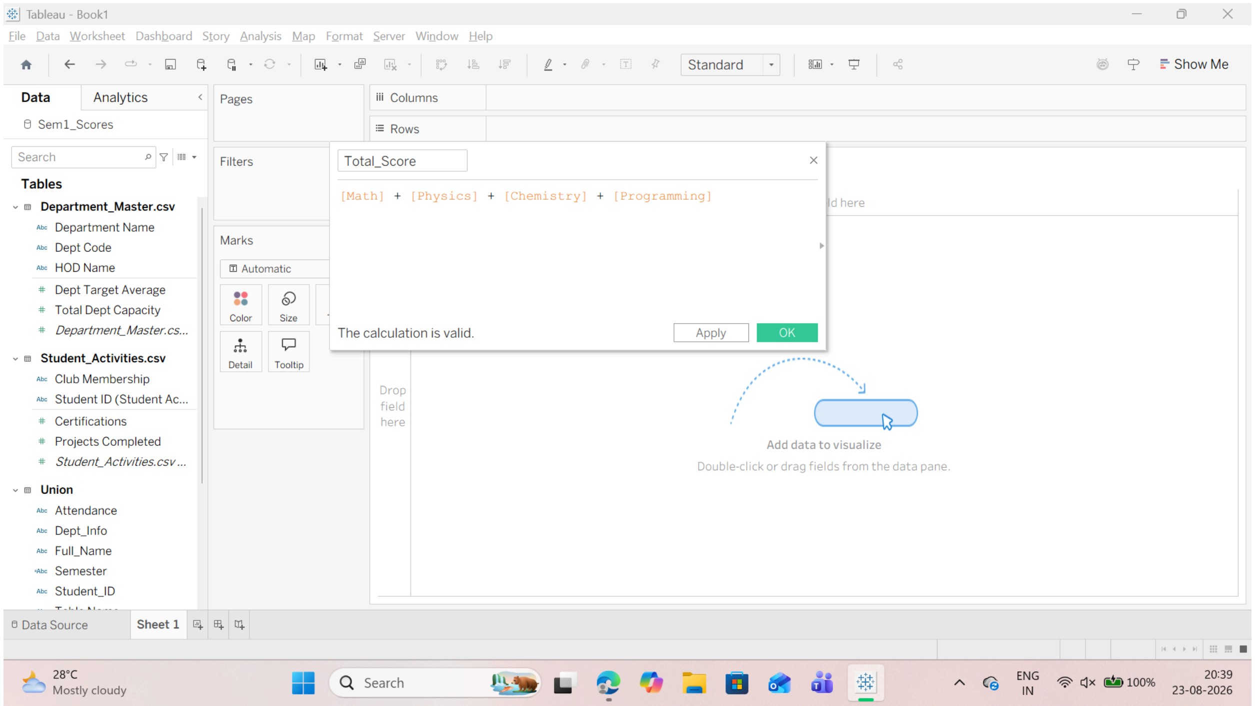
Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the student's total marks field:

o Field Name: Total_Score

o Tableau Formula:

[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]



4. Create the scaled overall percentage field:

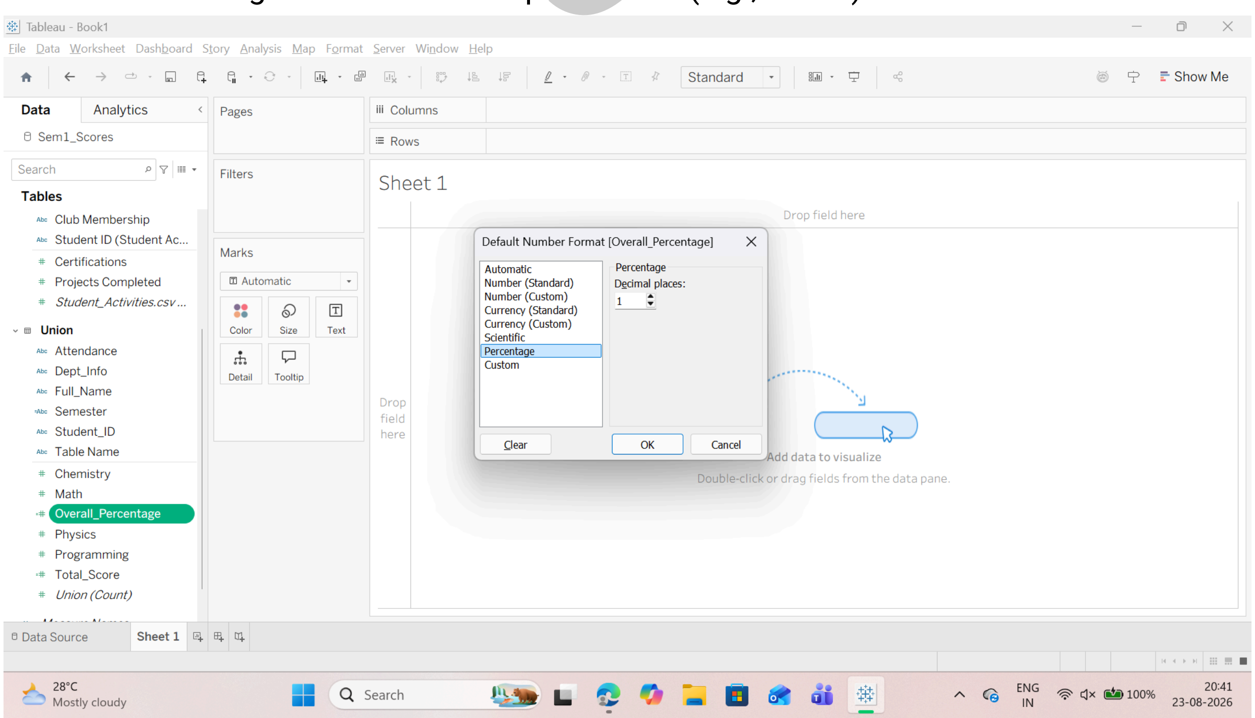
o Right-click in the Data Pane –> select Create Calculated Field....

o Field Name: Overall_Percentage o Tableau Formula: [Total_Score] / 400.0

5. Set the default percentage display formatting:

o In Data Pane, right-click Overall_Percentage >Default Properties>Number Format....

o Select Percentage and set decimal places to 1 (e.g., 85.5%).



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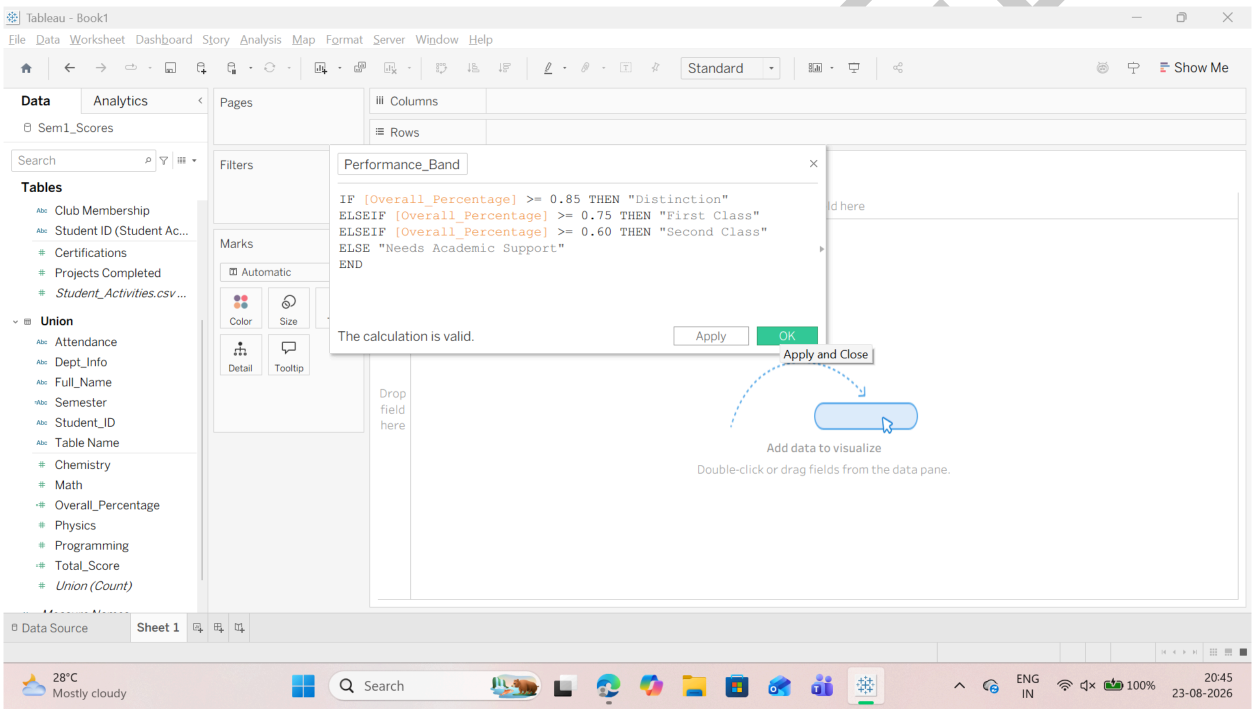
Step 2: Create Categorical Performance Bands (Grading Tiers)
1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create the academic grading tier field:
o Field Name: Performance_Band
o Tableau Formula:
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END



Step 3: Create an Attendance Health Flag
1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create the attendance compliance indicator:
o Field Name: Attendance_Status
o Tableau Formula:
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END

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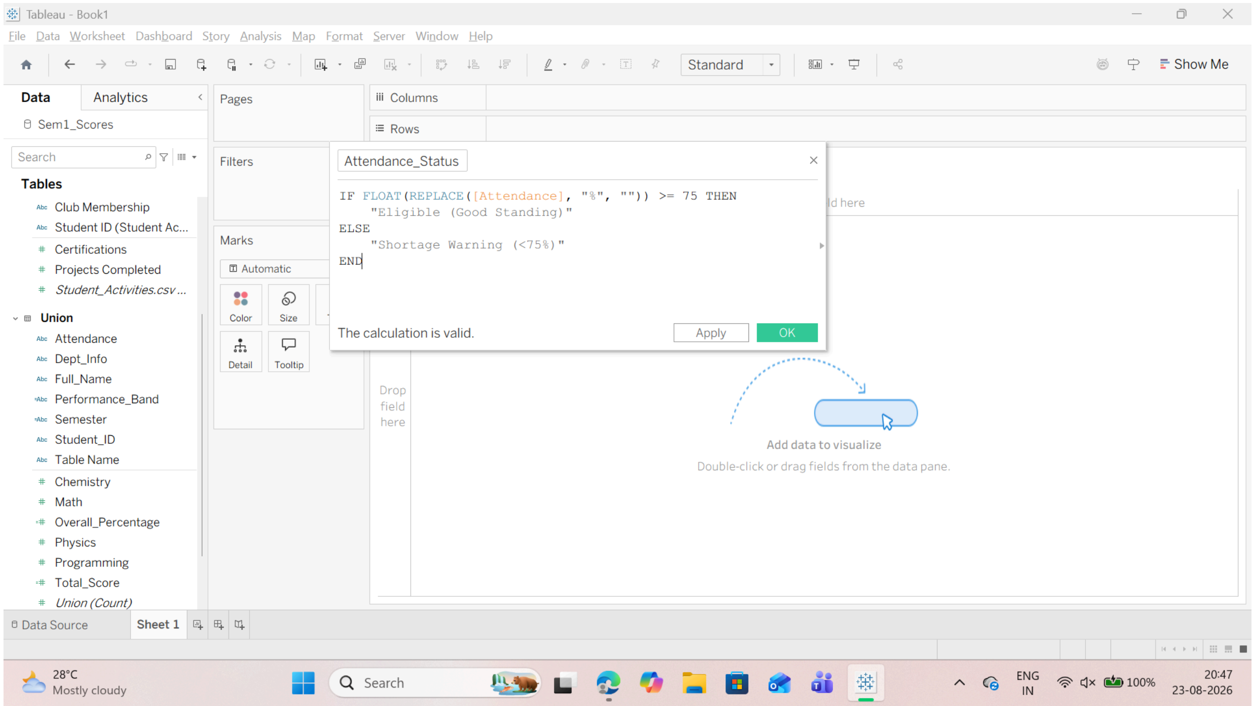
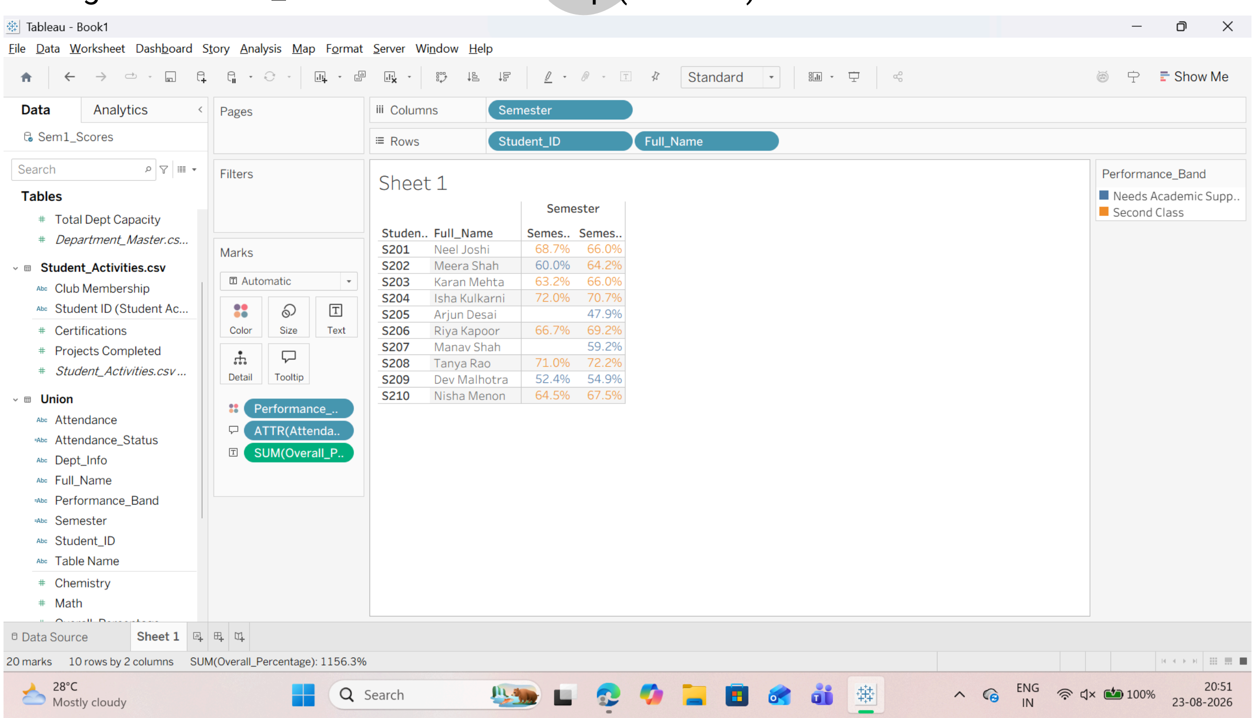


Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.



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F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the department-wide benchmark field:

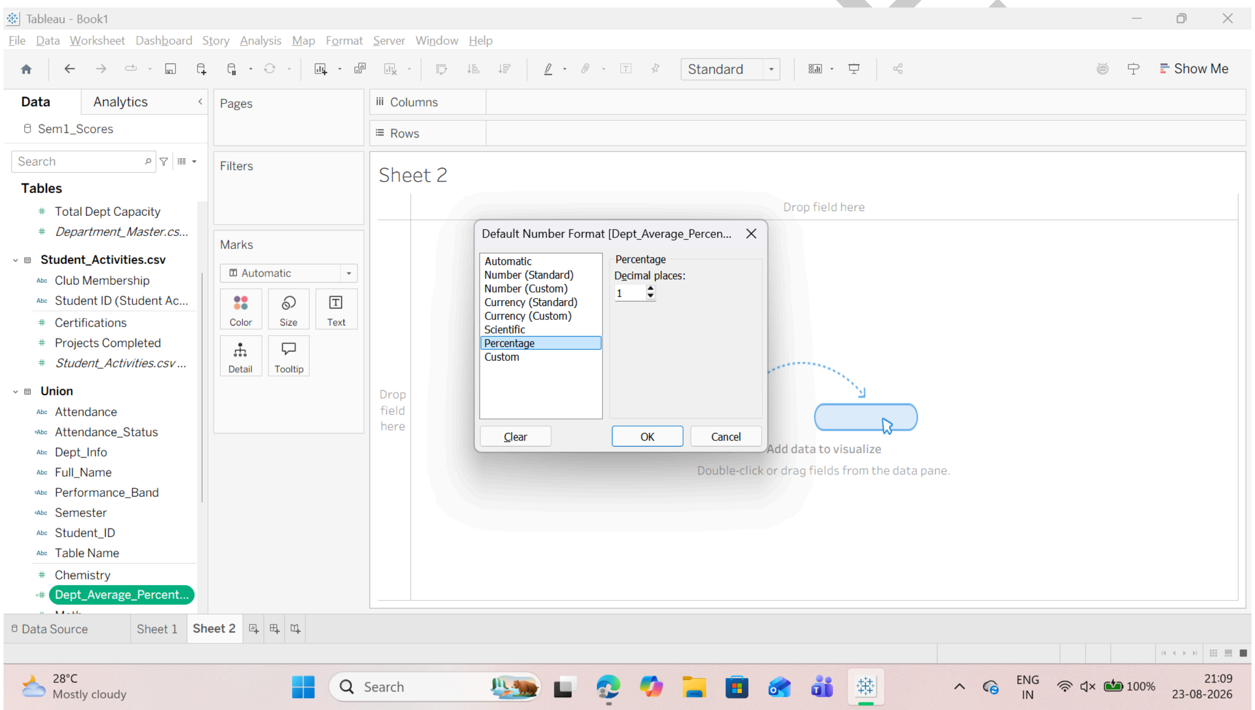
o Field Name: Dept_Average_Percentage_LOD

o Tableau Formula:

{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }

4. Set default formatting:

o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.



Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the cumulative student benchmark across all semesters:

o Field Name: Student_Overall_Career_Average

o Tableau Formula:

{ FIXED [Student_ID] : AVG([Overall_Percentage]) }

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

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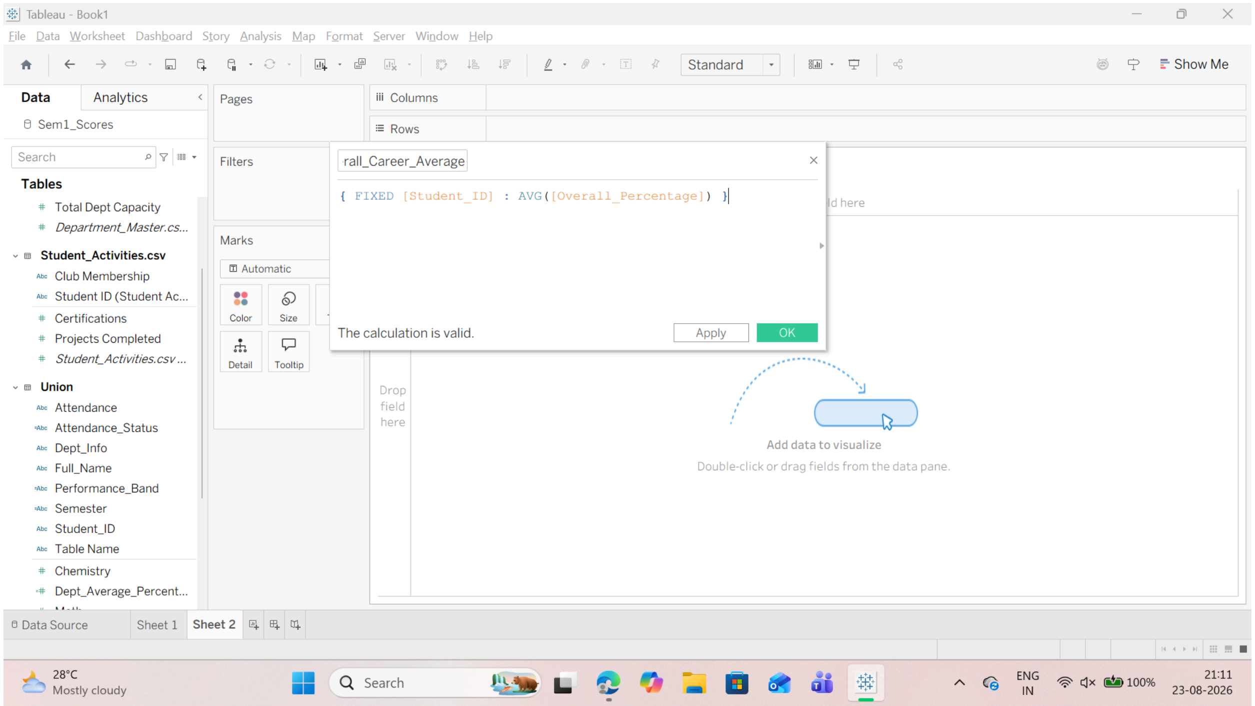


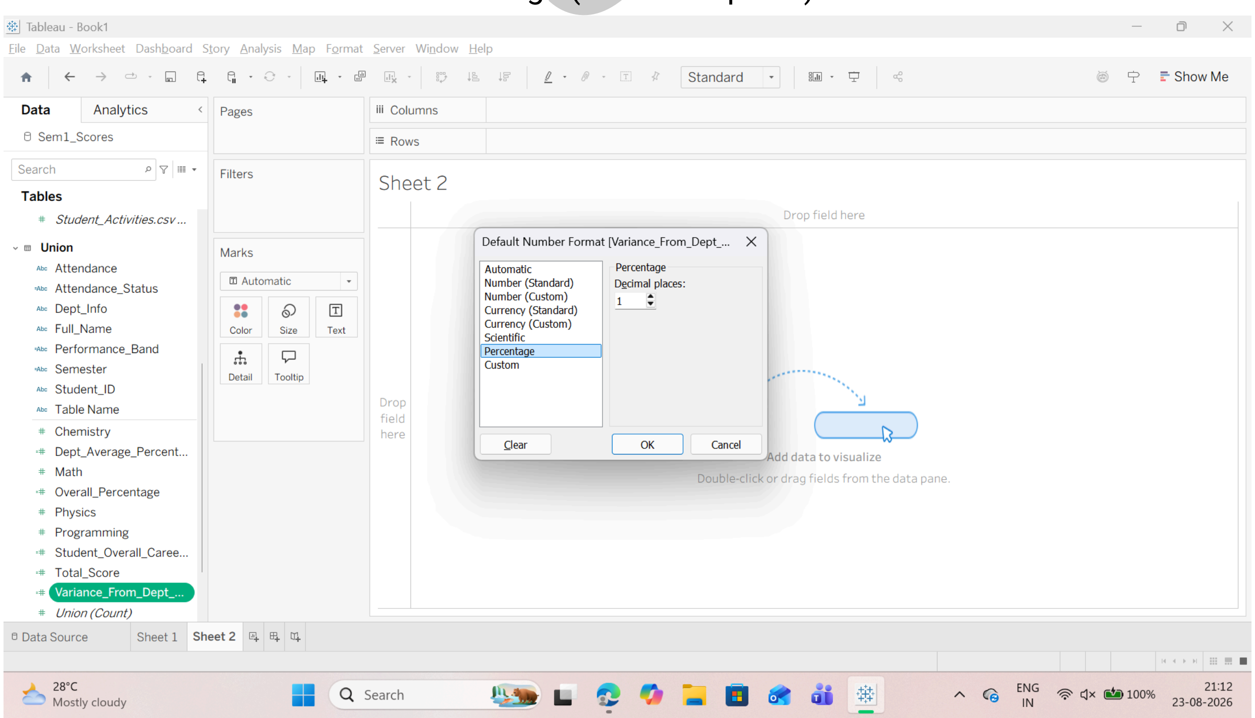
Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the relative difference metric:

o Field Name: Variance_From_Dept_Average

o Tableau Formula: [Overall_Percentage] - [Dept_Average_Percentage_LOD]

3. Set its Number Format to Percentage (1 decimal place).



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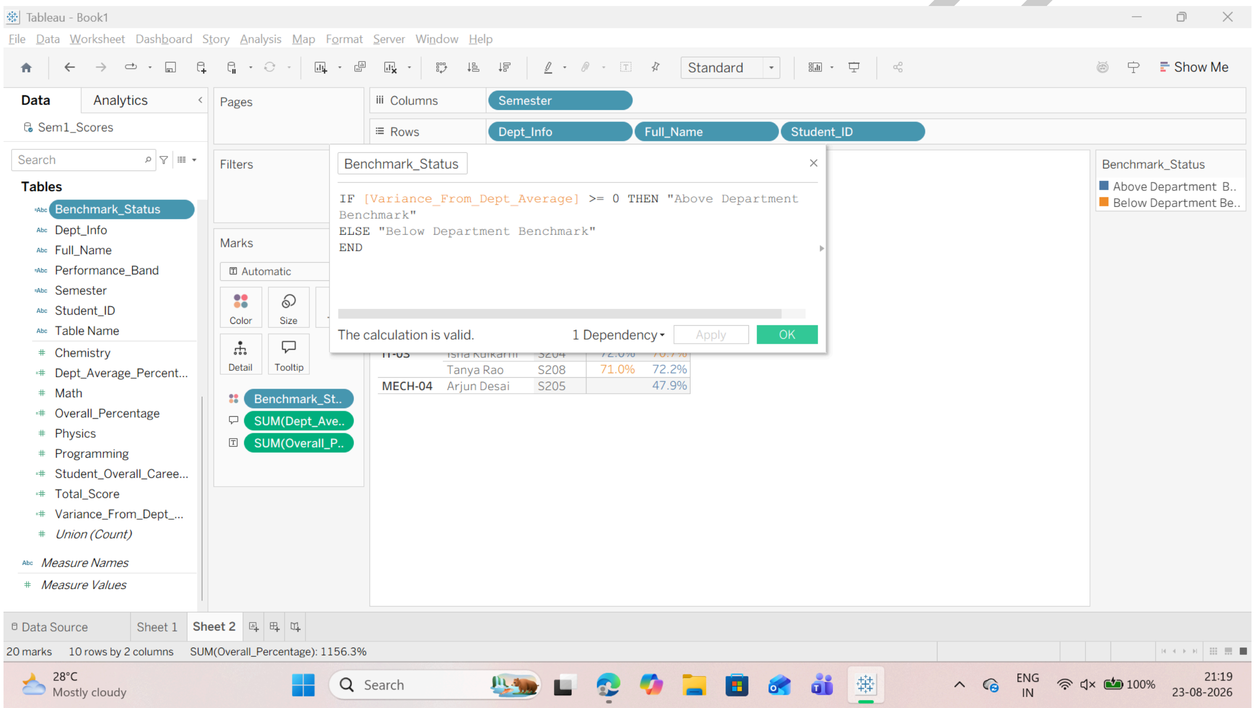
Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create an automated comparison label:

o Field Name: Benchmark_Status

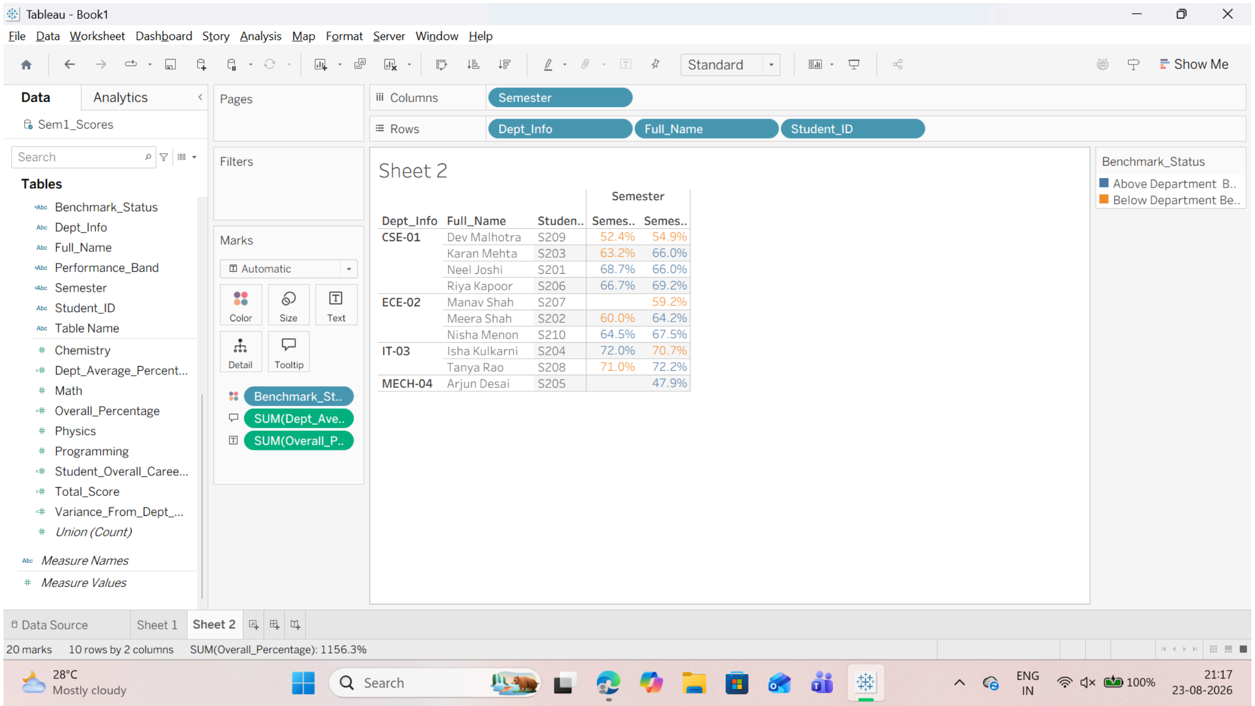
o Tableau Formula:

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END



Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.



G: Apply Advanced Data Cleaning Techniques
Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing
1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:
o Field Name: Full_Name_Clean
o Tableau Formula:
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,
1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,
LEN(TRIM([Full_Name]))))

o (Alternatively, for standard trim-only cleaning):
Code snippet
TRIM([Full_Name])
o Click OK.

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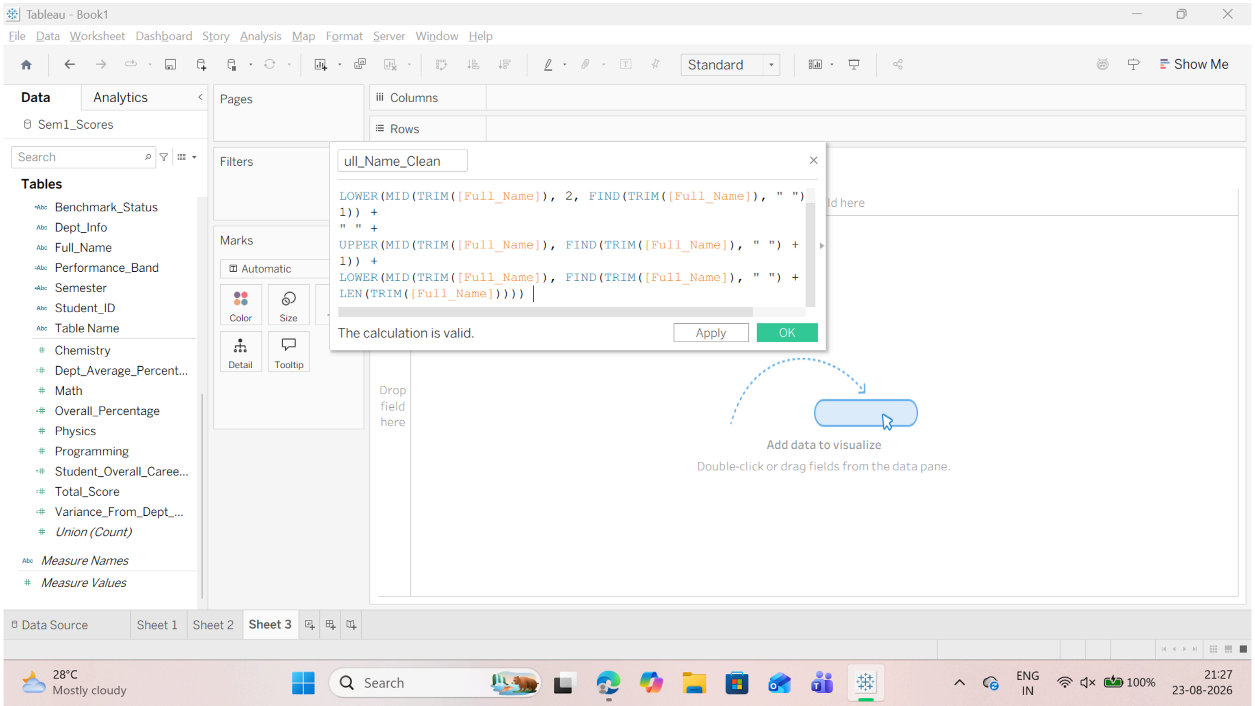


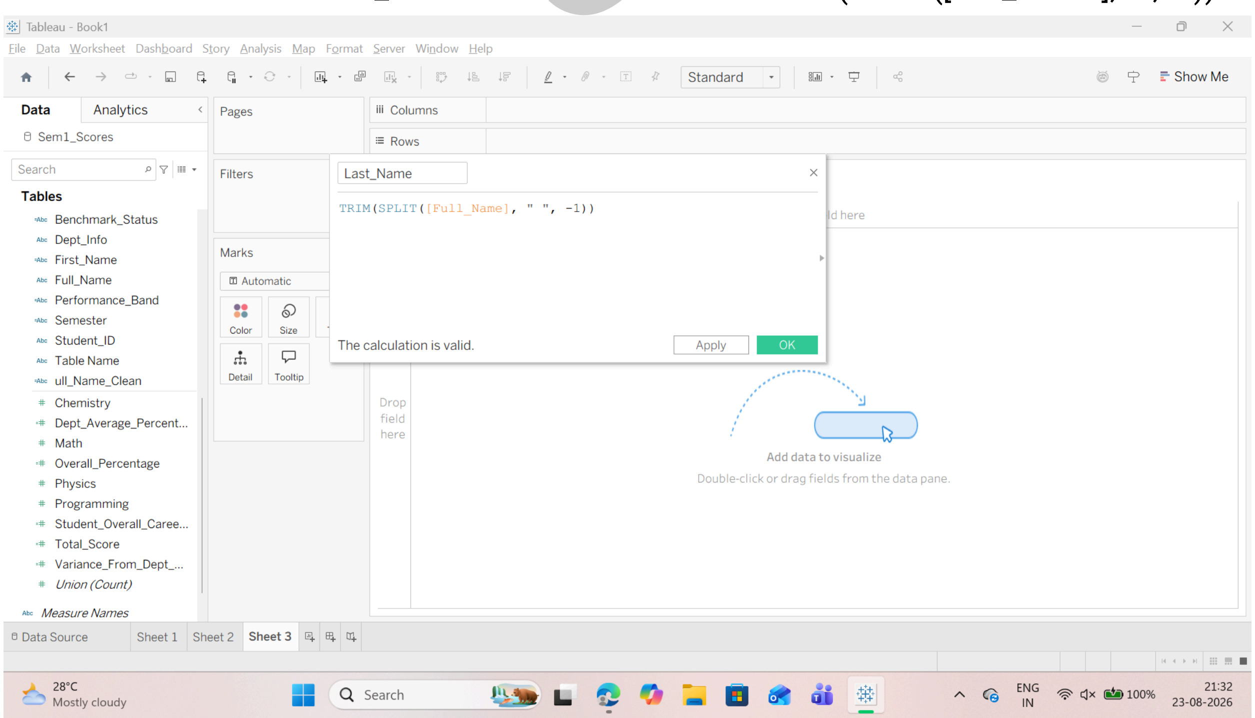
Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: First_Name o Tableau Formula: TRIM(SPLIT([Full_Name], " ", 1)) .

2. Extract the Last Name: o Right-click in the Data Pane --> select Create Calculated Field.... o Field Name: Last_Name o Tableau Formula: TRIM(SPLIT([Full_Name], " ", -1))



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Step 3: Split Composite Department Codes (Dept_Info)

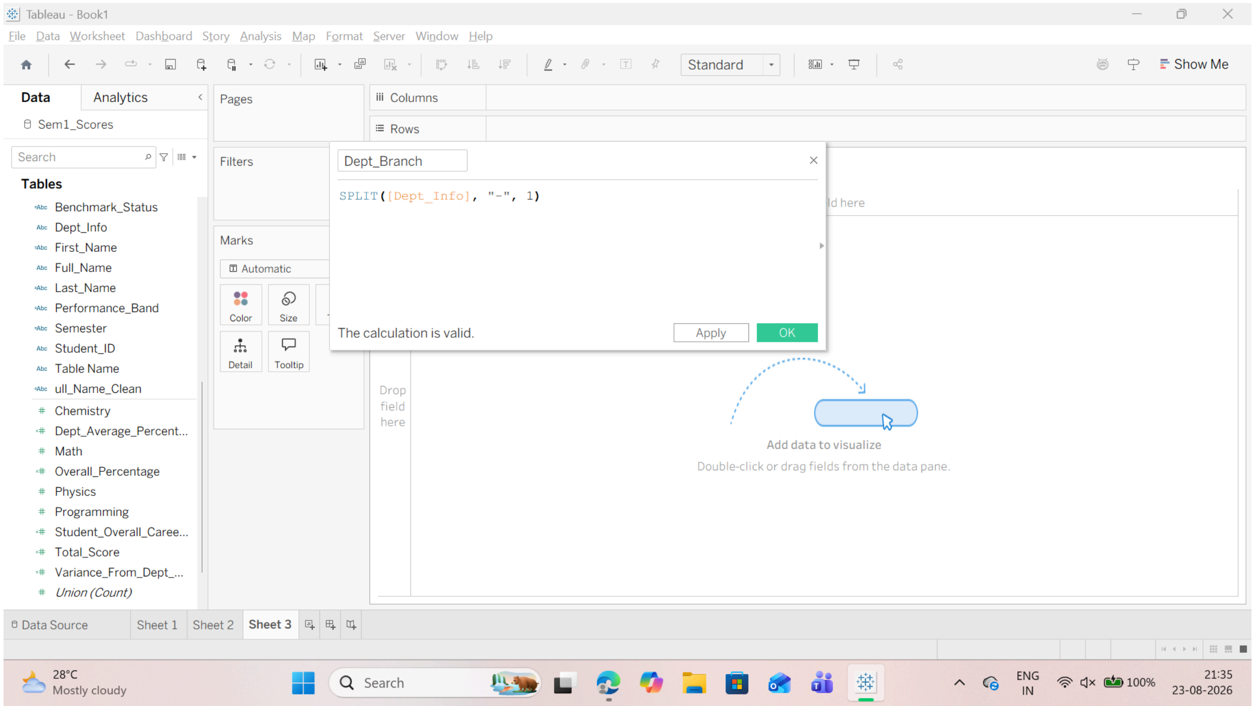
1. Extract the Department Acronym: 2. Extract the Batch / Section ID:

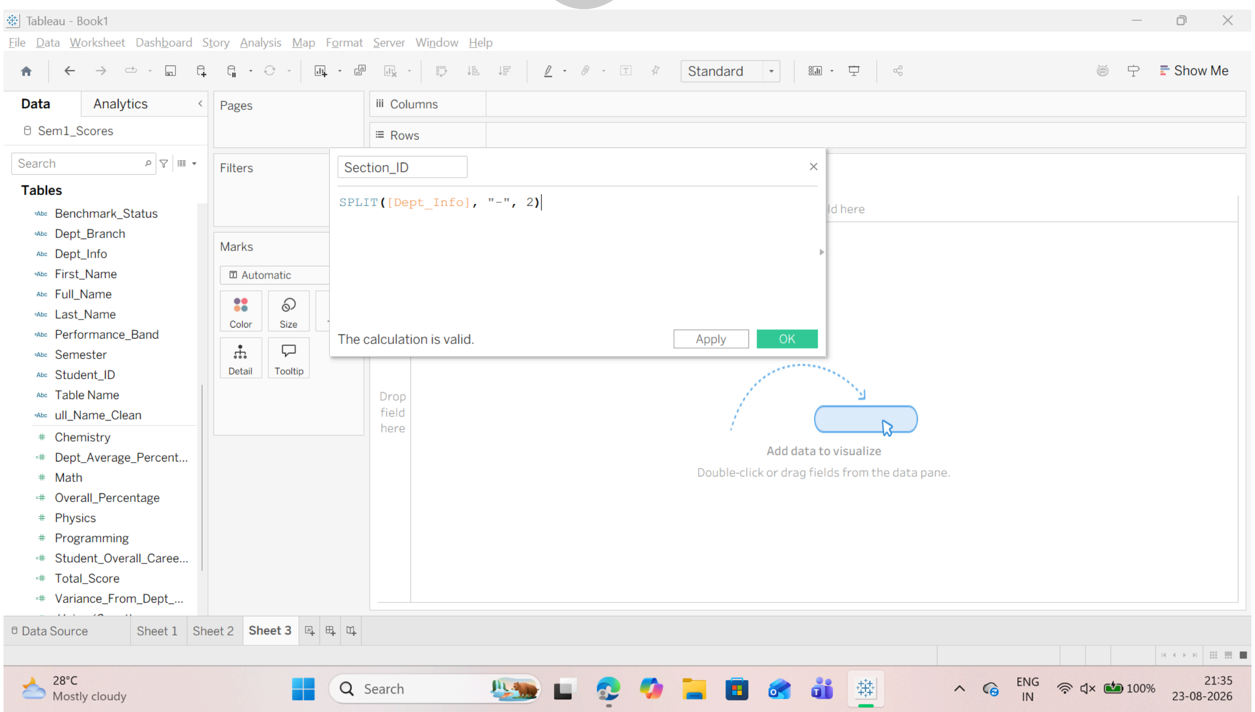
o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Dept_Branch o Tableau Formula: SPLIT([Dept_Info], "-", 1)

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Section_ID o Tableau Formula: SPLIT([Dept_Info], "-", 2)





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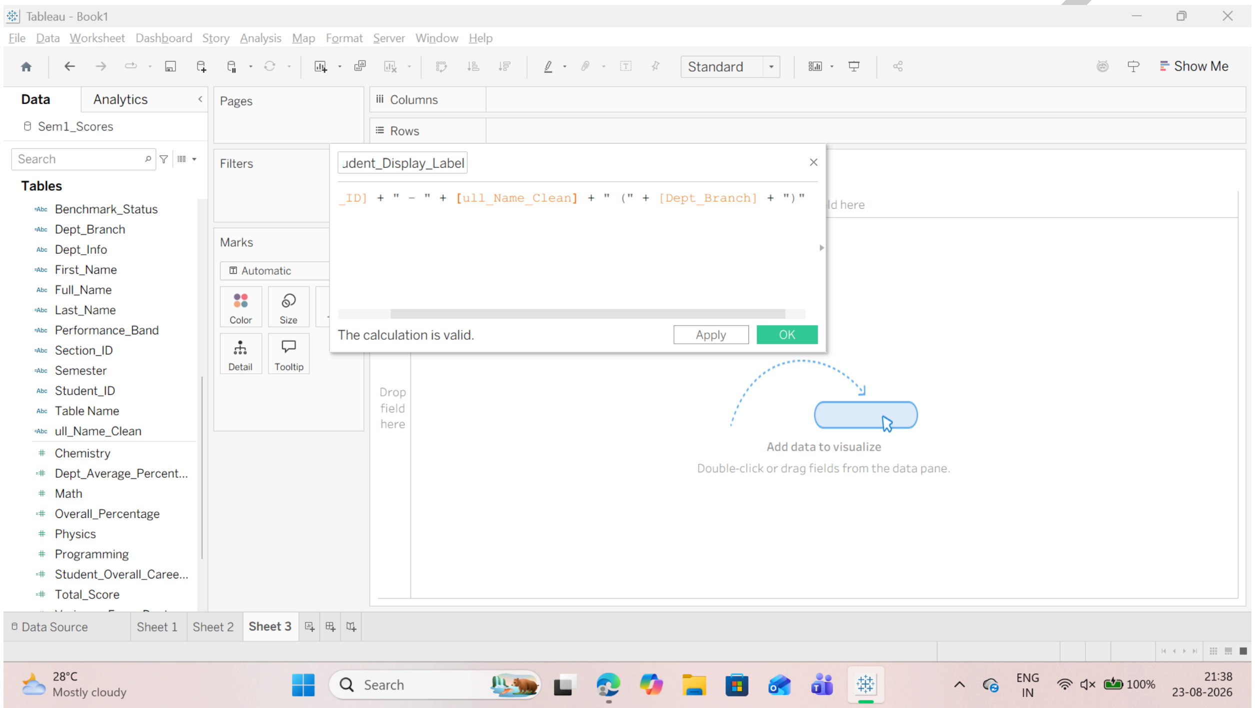
Step 4: Create a Clean Composite Student Label

Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Field Name: Student_Display_Label
3. Tableau Formula:

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"



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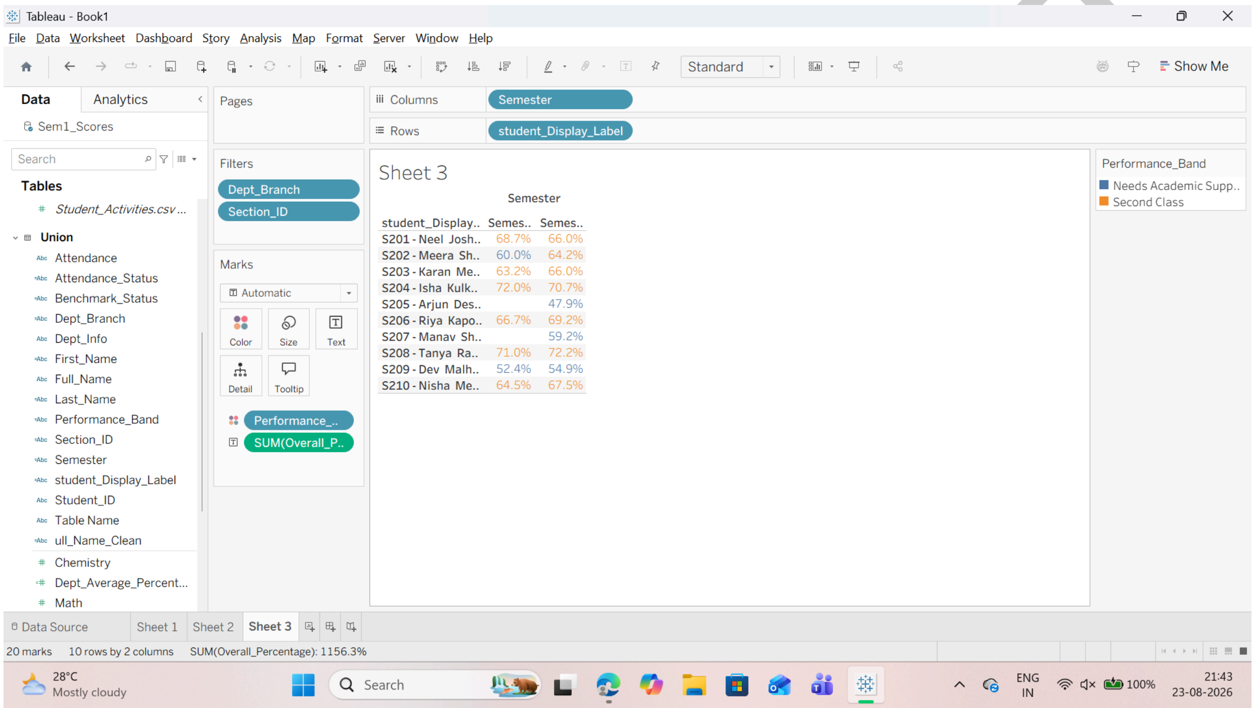
Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)



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